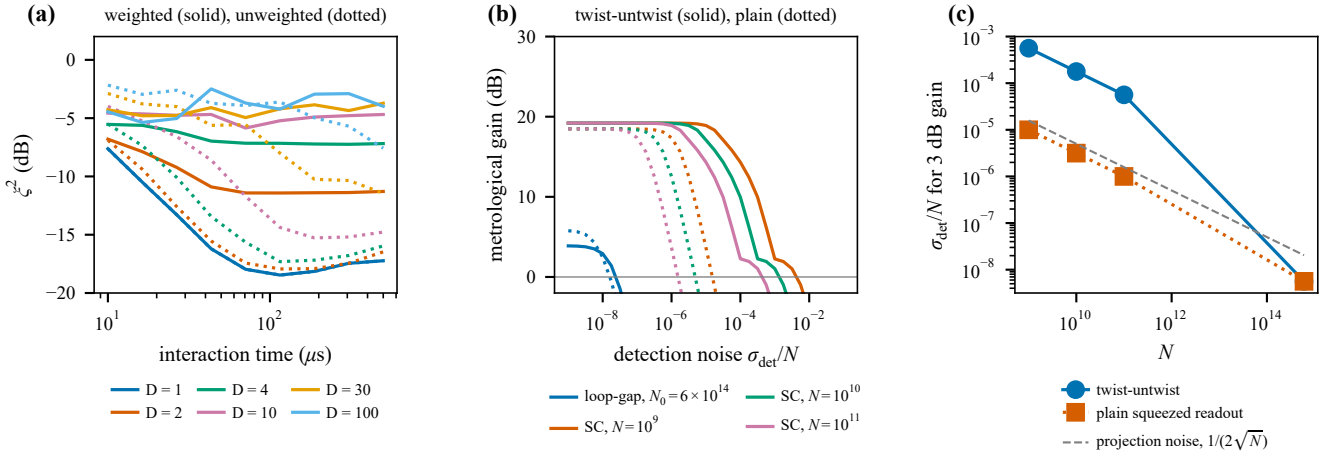
(a)
weighted (solid), unweighted (dotted)
ξ^2 (dB)
0
−5
−10
−15
−20
10^1
10^2
interaction time (μs)
D = 1
D = 2
D = 4
D = 10
D = 30
D = 100
(b)
twist-untwist (solid), plain (dotted)
metrological gain (dB)
30
20
10
0
10^{-8}
10^{-6}
10^{-4}
10^{-2}
detection noise σ_{det}/N
loop-gap, $N_0 = 6\times10^{14}$
SC, $N=10^9$
SC, $N=10^{10}$
SC, $N=10^{11}$
(c)
σ_{det}/N for 3 dB gain
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{10}
10^{12}
10^{14}
N
twist-untwist
plain squeezed readout
projection noise, $1/(2\sqrt{N})$